



ANYNET+ SIM CARD ES7XXX

Eseye's ES7xxx SIMs enable IoT devices to connect through mobile networks globally and switch networks to maintain optimum connectivity. A single SKU permits a device to deploy in all territories supported by the modem in use. Eseye supplies different IMSI configurations, represented by each x.

Features

- GSMA, ETSI TS 102 241 and ISO-7816, T=0 compliant
- eUICC and multi-IMSI
- Space for up to 10 profiles, remote profile management
- 4 logical channels
- Industrial grade
- Storage for application security parameters

Applications

- 2G, 3G, 4G/LTE and Cat M1 LTE IoT devices
- Global deployments
- Security
- Vending, Point of Sale terminals
- EV charge points
- Critical infrastructure monitoring
- Cloud-connected
- Edge processing devices
- Connected consumer applications

Standards

- 3GPP – Release 14
- Common Criteria – EAL 4+ (chipset)
- ETSI – TS 102 221 - UICC-Terminal interface; Physical and logical characteristics
- GlobalPlatform:
 - Card Specification Version 2.1.1
 - Card Specification Version 2.2 Amendments A, B, D & E
- GSMA:
 - SGP.01 Embedded SIM Remote Provisioning Architecture v1.1
 - SGP.02 Embedded UICC Technical Specification v3.2
 - SGP.16 M2M Compliance Process v1.1
- ISO/IEC 7816, T=0
- Oracle – Java Card 3 Platform, Classic Edition version 3.0.5
- REACH compliant
- RoHS compliant
- TCA – eUICC Profile Package: Interoperable Format Technical Specification v2.1

Device Information

Package	Delivered
MFF2	Reel
2FF/3FF/4FF	Full card
2FF/3FF/4FF	Half card

Pin Definitions

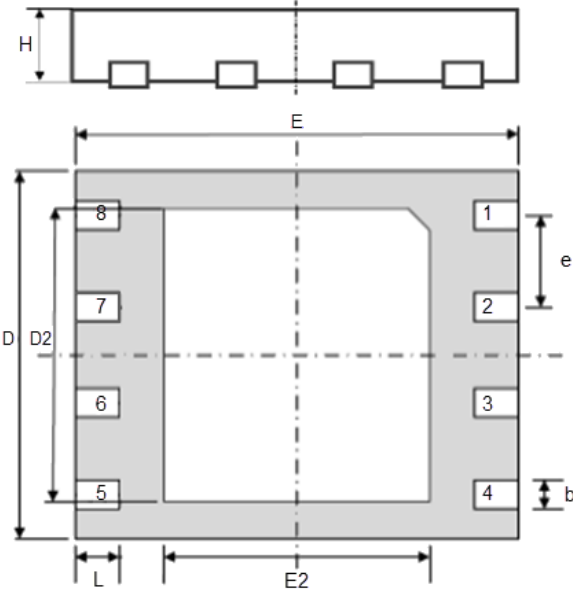
				
MFF2	2FF/ 3FF/ 4FF	Pin number	Pin name	Description
1	C5	1	Vss	Ground point
2	C6	2	n/c	Not connected
3	C7	3	I/O	In/Out [Data]
4		4	n/c	Not connected
5		5	n/c	Not connected
6	C3	6	CLK	Card Clock
7	C2	7	RST	Card Reset (Optional)
8	C1	8	VCC	Card Power (Supply voltage of the circuit)

*According to MFF2 specifications, pins 2, 4 & 5 are RFU. They should be floating (soldered onto the PCB but not connected to any signal).

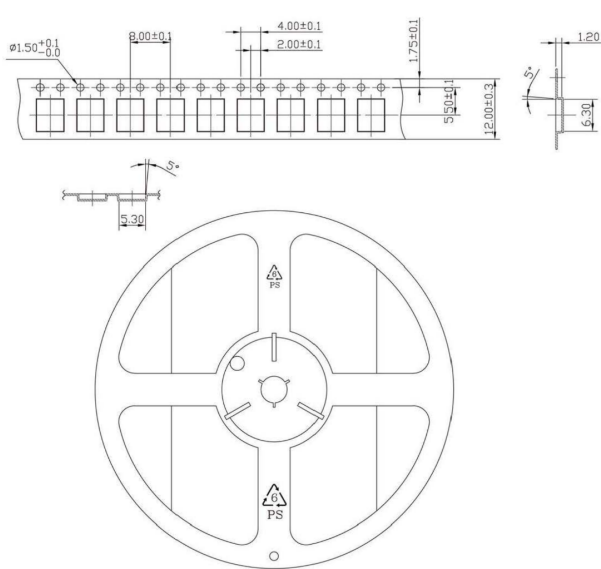
Operating parameters

	MFF2	2FF/3FF/4FF
Minimum operational and storage temperature (°C)	-40	-40
Maximum operational and storage temperature (°C)	+105	+105
Operating Voltage: Minimum, Maximum (V)	1.62, 5.5	1.62, 5.5
Plastic type	n/a	H-ABS (ABS 2.0)
Maximum reflow temperature (°C)	260	n/a
Write Endurance	500k per page	500k per page
Data retention time (years)	25	25
Minimum updates	10M	10M
Non-volatile memory (kb)	700 kB before OS loading; 375 kB after OS loading	700 kB before OS loading; 375 kB after OS loading
RAM (applets & NAA) (kb)	20 kB before OS loading; 10 kB after OS loading	20 kB before OS loading; 10 kB after OS loading
Data retention time at 85°C (years)	15	25

MFF2 package outline and footprint



MFF2 tape and reel dimensions



MFF2 tape and reel information

Tape feed : Single

Parameter	7" reel dimensions (mm)	13" reel dimensions (mm)
Units per reel	1000	3000
Reel diameter	178	330
Hub diameter	60	102
Tape width	12	12
Pocket pitch	8	8

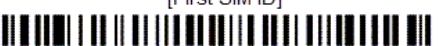
Pin #1 is identified by a cut or round corner in the central pad:

Extending the outer pad by 0.05-0.25mm from the package land helps improve the external solder joints.

MFF2 labels and quantity

A reel label is placed on each reel, inner packaging and inner box.

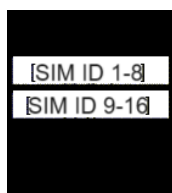
Format: 80 x 40 mm

Customer: ESEYE	
Order No: [OrderID]	Batch: [BatchNo]
Product: [ProductID]	Supplier: [SupplierID]
Date: [Date]	Plug Type: [FormFactor]
Qty box/total [X] / [XX] (box [Y] of [YY])	
From SIM ID:	
	[First SIM ID]
To SIM ID:	
	[Last SIM ID]

Embedded SIMs (MFF2) only

This information is lasered onto the back of the embedded SIM. Most customers will read the ICCID from the chip after installation.

[CODE] may be required for activation. Check product documentation for more information.



Items per reel: 1000 or 3000

Guidance on using MFF2

Solderability

- Au plating.
- Maximum reflow temperature: 260°C.
- Lead-free packaging compliant to the European Directive for Restriction of Hazardous Substances (RoHS).

Package handling

- Take standard preventative measures to prevent electrostatic discharge damage.
- Use appropriate handling processes to ensure the packages are free from moisture for soldering.



You can remount the MFF2 after it is soldered to a PCB.

PCB design recommendations

Solder and connect the central pad to the copper plane in order to improve attachment to the PCB and help heat dissipation.

For good solder joints and no bridging:

- PCB design complies with IPC-D-275.
- Flatness within 0.1mm per linear centimetre.
- Solderability complies with ANSI/J-STD-003.

If vias holes are required for maximised heat dissipation:

- Use typical dimensions for thermal vias: diameter 0.25-0.35mm, pitch 1.0-1.2mm.
- Plug vias to prevent solder wicking using tenting, Liquid Photo-Imageable (LPI) solder mask, or encroaching.

To prevent spikes on the supply voltage pins:

- Choose an appropriate decoupling capacitor (100nF is commonly used).
- Place capacitor between GND (VSS) and VCC (VDD), as close as possible to the VCC pin.

Mounting recommendations

Component placement and alignment

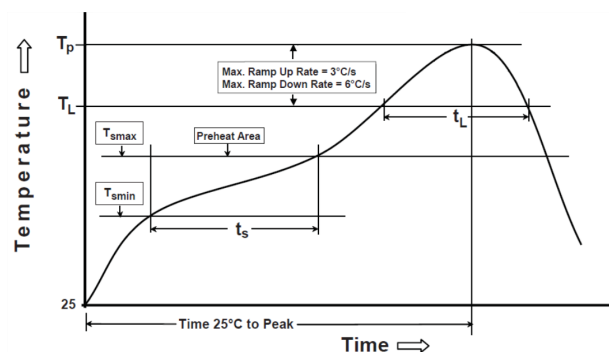
- To prevent shorting from solder bridging, ensure that no component is misaligned by more than 50% from pad centre.

Solder paste

- Use a no-clean solder paste if possible, although a lead-free solder paste metal alloy composition can be used. We recommend Type 3 paste, J-STD-005.
- Ensure solder paste is suitable for printing the solder stencil aperture dimensions.

Reflow

- All standard reflow soldering processes and typical temperature profiles are suitable for MFF2 board assembly, including:
 - Forced convection (recommended)
 - Vapor phase
 - Infrared
- Wave soldering is not possible.
- When using infrared ovens without convection, carefully ensure a sufficiently homogeneous temperature profile for all solder joints on the PCB, especially on large, complex boards with different thermal masses of the components. This includes the boards under the MFF2.
- Pb-Free Assembly.



Parameter	Amount
Preheat/Soak:	
T_{smin} – Minimum temperature	150°C
T_{smax} – Maximum temperature	200°C
t_s – Time in seconds	60-120s
T_p – Peak temperature	260°C
T_L – Liquidous temperature	5°C/second max
t_L – Time	60-120s
Time 25°C to peak temperature	8 minutes max

PCB cleaning

- If a no-clean solder paste and flux are used, PCB cleaning is usually not required. However, some types of no-clean solder paste may contain some contamination on the final board. Check and remove any residue on the board.

Solder joint inspection

- Use 2D or 3D techniques to measure the volume of solder plated on the PCB.
- Use Automated Optical Inspection (AOI) or an x-ray inspection system to check for defects after re-flowing the mounted package.
- Use cross-sectioning to verify fillet shape, size, and joint standoff height.

Ordering Information

Package	Description
MFF2	Embedded AnyNet+ SIM
2FF/3FF/4FF	AnyNet+ SIM Triple Cut
2FF/3FF/4FF	AnyNet+ SIM Half Card

Technical Support:

UK (Head office) & EU: +44 1483 802503
 USA: +1 484-935-3130
 Australia: +61 8 9551 5200
 Brazil: +55 11 4950-7015
 Turkey: +90-212-275-2626

Email: support@eseye.com

Sales:

UK (Head office) & EU: +44 1483 802501
 USA: +1 512-813-0599
 Africa: +27 87 551 8200
 Brazil: +55 11 5059-1574
 Turkey: +90-212-275-2626

Email: sales@eseye.com